ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5]

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Warning

- **If the cooling system cap or the water-cooled charge air cooler reserve tank cap is removed when the engine is hot, steam and boiling water may be injected, causing severe burns or severe injury. Perform the removal of the cooling system cap and water-cooled charge air cooler reserve tank cap when the engine is cool.**
- **Turn off the engine and wait until it is cool. Even then, be very careful when removing the cap. Wrap a thick cloth around it and slowly turn it counterclockwise to the first stop. Step back while the pressure escapes.**
- **When you are sure all the pressure is gone, press down on the cap using the cloth, turn it, and remove it.**

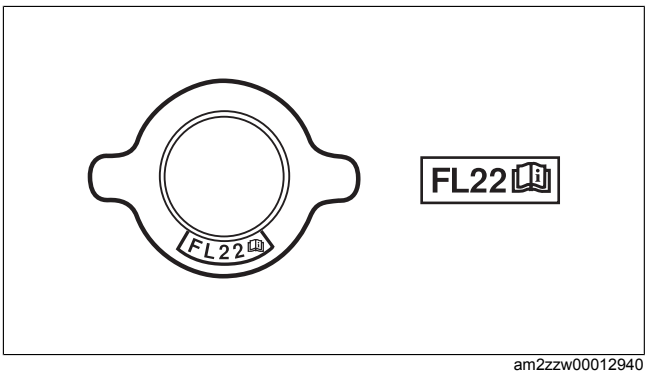
Caution

- **Use engine coolant at a concentration that meets the environmental conditions in which the vehicle is driven, otherwise engine damage could occur.**
- **The engine has aluminum parts and must be protected by an ethylene-glycol-based coolant to prevent corrosion and freezing.**
- **Do not use coolants containing Alcohol, Methanol, Borate or Silicate. These coolants could damage the cooling system.**
- **Use only soft (demineralized) water in the coolant mixture. Water that contains minerals will cut down on the coolant's effectiveness.**
- **Engine coolant damages paint. If engine coolant does get on a painted surface, rinse it off quickly.**

Note

- If the “FL22” mark is shown on or near the cooling system cap, use FL22 type engine coolant.
- FL22 type engine coolant is shipped as a diluted solution. Use the solution as is when replacing coolant.

Engine coolant capacity (approx. quantity)
MTX: 6.6 L {7.0 US qt, 5.8 Imp qt}
ATX (Israel): 6.7 L {7.1 US qt, 5.9 Imp qt}
ATX (Except Israel): 6.8 L {7.2 US qt, 6.0 Imp qt}

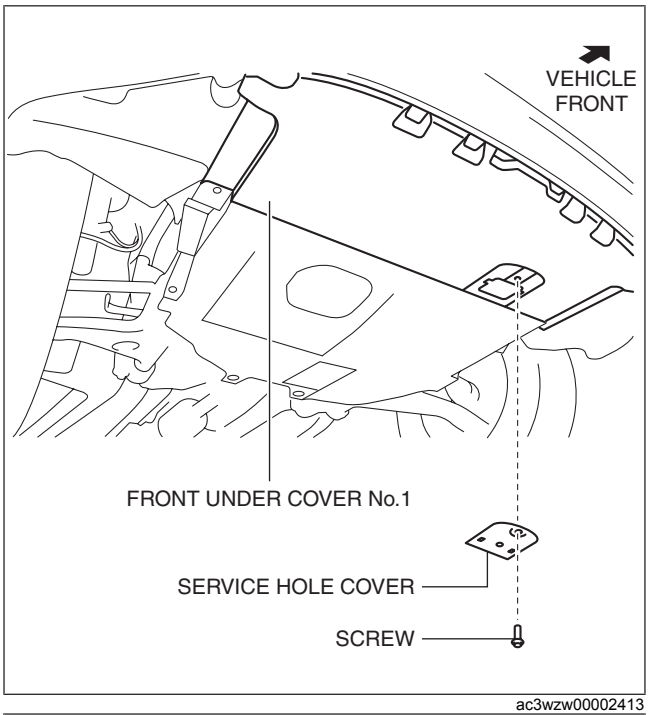


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Engine Coolant Replacement Procedure

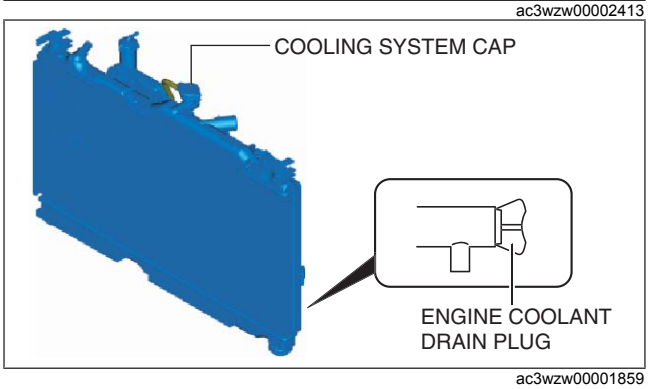
1. Remove the cooling system cap.
2. Drain the engine coolant from the coolant reserve tank.

3. Remove the service hole cover (installed to front under cover No.1) used to drain the engine coolant.



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4. Loosen the engine coolant drain plug and drain the engine coolant into a container.
5. After completely draining the engine coolant, tighten the radiator drain plug.
6. Referring to the following chart, select the correct volume percentage of the water and engine coolant.



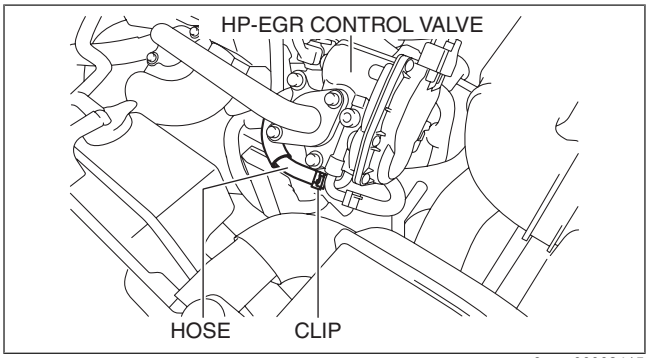
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Antifreeze solution mixture percentage (Except FL22 type engine coolant)

Engine coolant protection	Volume percentage (%)		Gravity at 20 °C {68 °F}
	Water	Coolant	
Above -16 °C {3 °F}	65	35	1.057
Above -26 °C {-15 °F}	55	45	1.072
Above -40 °C {-40 °F}	45	55	1.086

7. Disconnect the hose shown in the figure from the HP-EGR control valve.

Caution
- **Cover the area around the hose with a rag so that engine coolant leaking from the hose does not get on the surrounding parts.**



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8. Add engine coolant to the radiator from the radiator cap opening until it begins to leak from the disconnected hose.
9. Connect the HP-EGR control valve to the hose.
10. Refill the engine coolant from the radiator filler neck until it is close to the top of the radiator filler neck.
11. Refill the engine coolant into the coolant reserve tank up to the FULL mark on the tank.
12. Install the cooling system cap.
13. Perform air bleeding. (See Air Bleeding Procedure.)
14. Install the service hole cover.

Air Bleeding Procedure

1. Perform air bleeding using the following procedures.

Note
- If the accelerator pedal is depressed continuously for a specified time, the engine speed may decrease to the idle speed. This is due to the fuel cut control operation, which prevents overheating, and it does not indicate a malfunction.

(1) Run the engine at **2,500 rpm** for 5 min.
(2) Maintain the engine speed at **3,000 rpm** for 5 s, and then idle.
(3) Repeat Steps (1), (2) several times.

2. Stop the engine and wait until the engine coolant temperature decreases.
3. Inspect the engine coolant level.
   - If engine coolant level is low, add engine coolant and perform air bleeding again.
4. Inspect for engine coolant leakage. (See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)